

MAT 534: Algebra I

Aditya Pahuja

Contents

1	August 25	2
2	August 27	5
3	September 3	7
4	September 8	8

1 August 25

Definition 1.1 (Group) — A **group** is a set G with a binary operation

$$\begin{aligned} *: G \times G &\rightarrow G \\ (g, h) &\mapsto g * h \end{aligned}$$

such that

- (1) $*$ is associative, so $f * (g * h) = (f * g) * h$.
- (2) There is a **neutral element** $e \in G$ such that for all $g \in G$, $e * g = g * e = g$.
- (3) For all $g \in G$, there is an **inverse element** $h \in G$ such that $g * h = h * g = e$.

Some basic facts:

- e is unique, because if e and e' are neutral then $e = e * e' = e'$.
- For each $g \in G$, the inverse of g is unique, for if h and h' are inverses of g then $h = h * (g * h') = (h * g) * h' = h' = h'$. In a generic group we write g^{-1} for the inverse of g .
- The inverse of g^{-1} is g .

From here on, when working with a generic group, we use multiplicative notation, meaning

- multiplication is written either with juxtaposition or, if emphasis is need, a multiplication dot: $g * h = gh = g \cdot h$.
- repeated multiplication is given by exponents: if n is a positive integer, then g^n is the n -fold product of g ; if $n = 0$, then $g^0 = 1$, and if n is negative, then $g^n = (g^{-1})^{-n}$, the $-n$ -fold product of g^{-1} .

Definition 1.2 (Abelian group) — G is **abelian** if its multiplication operation is commutative, i.e. $gh = hg$ for all $g, h \in G$.

For generic abelian groups we instead prefer additive notation, so

- we write $g + h$ instead of gh or $g \cdot h$ or $g * h$ for the operation;
- we write 0 instead of 1 for the neutral element;
- we write $-g$ instead of g^{-1} for inverses;
- we write ng instead of g^n for n -fold applications of the operation where n is an integer.

Example 1.3 (Examples of groups)

Some abelian groups are the additive groups $(\mathbb{Z}, +)$, $(\mathbb{Q}, +)$, $(\mathbb{R}, +)$, $(\mathbb{C}, +)$, ... and the multiplicative groups $(\mathbb{Q}^\times, \cdot)$, $(\mathbb{R}^\times, \cdot)$, etc. where the superscript $\times$ means “nonzero” (so $\mathbb{Q}^\times$ is the set of nonzero rationals.)

In real life groups often show up as symmetries of objects (where “symmetry” is often interpreted as “structure-preserving,” such as, say, the group of isomorphisms between two groups, the group of automorphisms on an algebraic extension of $\mathbb{Q}$, the group of deck transformations on a covering space, etc.) A fairly literal example of this is the **dihedral group** D_{2n} , which is the group of symmetries of a regular n -gon, i.e. the set of rigid motions in space that map an n -gon onto itself, like rotations by $\frac{2\pi}{n}$ about the center or reflections over certain lines; the group operation is performing these rigid motions one after the other.

Exercise 1.4. Convince yourself that these transformations form a group.

To abstract this into a more symbolic form, suppose that the vertices of the n -gon are the points $e^{2\pi k/n}$ for $k = 1, \dots, n$ in the complex plane, so that the real line is an axis of symmetry of the n -gon. Then, let r represent rotation of the n -gon counterclockwise by $\frac{2\pi}{n}$ and d reflection over the real axis.

Exercise 1.5. Show that $drd = r^{-1}$, and hence that the elements of D_{2n} are the powers of r ($1, r, \dots, r^{n-1}$) and d times these powers of r ($d, dr, \dots, dr^{n-1}$). This means that D_{2n} has $2n$ elements, justifying the subscript.

We can also express D_{2n} with a **group presentation**, namely

$$D_{2n} = \langle r, d \mid r^n = 1, d^2 = 1, drd = r^{-1} \rangle;$$

this says that D_{2n} is the group with two generators r, d that can be algebraically manipulated with the given relations. This is in general not that fruitful of a task, though, since the problem of deciding whether two elements of a group are equal by looking at the group’s presentation is undecidable.

Another example is the **matrix group** $GL_n(F)$, which is the group of invertible $n \times n$ matrices whose entries are in the field F (this of course consists of the $n \times n$ matrices A with nonzero determinant), whose operation is matrix multiplication. Checking that this is a group:

- (1) Matrix multiplication is associative because matrices in $GL_n(F)$ are just concrete representations of linear maps on F^n , and function composition is associative.
- (2) The neutral element is the identity matrix

$$I = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}.$$

- (3) The inverse of a matrix can either be computed concretely or, again understanding matrices to be linear maps, one can directly construct the inverse of the corresponding linear map by looking at bases of the vector spaces that the map acts on.

A final very useful group is the **symmetric group** S_n , which is the group of **permutations** on a set of n things (for concreteness, the n things are the elements of $[n] := \{1, \dots, n\}$); that is, S_n is the set of bijections $\sigma: [n] \rightarrow [n]$ endowed with the operation of function composition (thus multiplication happens right-to-left here; we did not need to worry about

this with matrices because we can imagine them as just grids of numbers for which we have an admittedly obtuse multiplication rule). Note that $|S_n| = n!$, since there are n choices for $\sigma(1)$, followed by $n - 1$ choices for $\sigma(2)$, and so on, for a total of $n \cdot (n - 1) \cdots 2 \cdot 1 = n!$.

When working with permutations, we would like to have a good way to write them down; naively, one can simply write down a table

i	1	2	3	4	5	6	7	8	9	10	11	12	13
$\sigma(i)$	12	13	3	1	11	9	5	10	6	4	7	8	2

but this is clunky and annoying, as well as not telling us much about the structure of σ (which is why we'd rather not just resort to a more compact 13-tuple).

Instead, we use **cycle notation**.

Definition 1.6 (Cycles) — Given distinct $a_1, \dots, a_k \in \{1, \dots, n\}$, define

$$(a_1 \ a_2 \ \dots \ a_k)$$

to be the permutation that maps $a_1 \mapsto a_2 \mapsto \dots \mapsto a_k \mapsto a_1$. We call this a **k -cycle**.

It is a theorem (that we will not prove rigorously) that any permutation in S_n can uniquely be expressed as a product $\sigma_1 \sigma_2 \cdots \sigma_r$ of pairwise disjoint nontrivial cycles σ_j , up to the ordering of the σ_j (pairwise disjoint every element of $[n]$ is fixed by all but one of the σ_j , and nontrivial meaning the cycles aren't the identity permutation, so not 1-cycles). The way that one computes this cycle representation of a permutation (and, incidentally, proves that this representation exists and is unique) is simply by following where the numbers are mapped to, say by using the table. For example, in the table of $\sigma \in S_{13}$ above,

- $1 \mapsto 12 \mapsto 8 \mapsto 10 \mapsto 4 \mapsto 1$, giving the cycle $(1 \ 12 \ 8 \ 10 \ 4)$.
- $2 \mapsto 13 \mapsto 2$, giving the cycle $(2 \ 13)$.
- 3 maps to itself, giving the cycle (3) , which we omit.
- $5 \mapsto 11 \mapsto 7$, giving the cycle $(5 \ 11 \ 7)$.
- $6 \mapsto 9 \mapsto 6$, giving the cycle $(6 \ 9)$.

Therefore,

$$\sigma = (1 \ 12 \ 8 \ 10 \ 4)(2 \ 13)(5 \ 11 \ 7)(6 \ 9).$$

As another example of cycle notation, we can write

$$S_3 = \{(1), (1 \ 2), (1 \ 3), (2 \ 3), (1 \ 2 \ 3), (1 \ 3 \ 2)\}$$

where by convention the identity permutation is written as (1) because the empty string is hard to write. Recalling that composition of permutations happens right-to-left, we can compute

$$(1 \ 2)(1 \ 3) = (1 \ 3 \ 2)$$

by following where each element goes (imagine moving the numbers around in a line): 1 goes to position 3, and then stays there because $(1 \ 2)$ doesn't touch position 3; the newly displaced 3 goes to position 1 and then is promptly moved to position 2; and finally 2, which is not moved by $(1 \ 3)$, is sent to position 1, so the overall map is $1 \mapsto 3 \mapsto 2 \mapsto 1$. On the other hand, $(1 \ 3)(1 \ 2) = (1 \ 2 \ 3)$ by the same logic, so in fact we have shown that S_3 (and generally S_n) is non-abelian for all $n \geq 3$.

2 August 27

Definition 2.1 (Homomorphism) — Let G and H be groups. A **homomorphism** $\varphi: G \rightarrow H$ is a function such that

$$\varphi(g_1 g_2) = \varphi(g_1) \varphi(g_2).$$

If φ is bijective, it is called an **isomorphism** and G, H are said to be **isomorphic**, which is denoted as $G \cong H$.

Some simple facts:

- $\varphi(1_G) = 1_H$ because $\varphi(1_G) = \varphi(1_G) \cdot \varphi(1_G)$.
- $\varphi(g^{-1}) = \varphi(g)^{-1}$ because $\varphi(g) \varphi(g^{-1}) = \varphi(1_G) = 1_H$.
- If $\varphi: G \rightarrow H$ is an isomorphism then so is $\varphi^{-1}: H \rightarrow G$.

In fact, isomorphism is an equivalence relation on groups.

Example 2.2

Some examples of homomorphisms:

- (1) The determinant $\det: GL_n(F) \rightarrow F^\times$ is multiplicative, so it defines a homomorphism on the set of invertible linear maps.
- (2) If the vertices of a regular n -gon are labeled $1, \dots, n$, then D_{2n} permutes these vertices; therefore we get a homomorphism $D_{2n} \rightarrow S_n$ by sending $d \mapsto (1)(2 \ n)(3 \ n-1) \cdots$ and $r \mapsto (1 \ 2 \ \cdots \ n)$.
- (3) The groups $(\mathbb{R}, +)$ and $(\mathbb{R}_{>0}, \cdot)$ are isomorphic, with isomorphism $\exp: \mathbb{R} \rightarrow \mathbb{R}_{>0}$.
- (4) Fix $g \in G$. Then, the **conjugation map** is an isomorphism $\varphi_g: G \rightarrow G$ given by $\varphi_g(x) = gxg^{-1}$. Moreover, $\varphi_a \varphi_b = \varphi_{ab}$, so the group $\{\varphi_g : g \in G\}$ with operation being group composition is isomorphic to G .

Definition 2.3 (Subgroup) — A **subgroup** of a group G is a subset $H \subseteq G$ that is a group under the operation of G ; in particular,

- (1) $1_G \in H$
- (2) $h_1, h_2 \in H \implies h_1 h_2 \in H$
- (3) $h \in H \implies h^{-1} \in H$.

We more succinctly write $H \leq G$ if H is a subgroup of G . Furthermore H is **normal** in G if $\varphi_g(H) = H$, i.e. $h \in H$ implies $ghg^{-1} \in H$; this is denoted $H \trianglelefteq G$.

Normal subgroups are a very important part of the structure of a group G due to the following proposition:

Proposition 2.4. If $\varphi: G \rightarrow H$ is a homomorphism then $\varphi(G)$ is a subgroup of H and the kernel $\ker \varphi = \{g \in G : \varphi(g) = 1_H\}$ is a normal subgroup of G .

Proof. Trivial; omitted. $\square$

Example 2.5

Some examples:

- (1) $SL_n(F)$, the subset of $GL_n(F)$ whose determinant is 1, is the kernel of the determinant homomorphism, so it is normal in $GL_n(F)$.
- (2) The **center** $Z(G) = \{g \in G : gh = hg \forall h \in G\}$ is a normal subgroup of G .

Proposition 2.6. The intersection of an arbitrary collection of subgroups of a group G is also a subgroup of G .

Proof. Trivial; omitted. $\square$

Definition 2.7 (Generated subgroup) — Let A be a subset of a group G . The subgroup $\langle A \rangle$ generated by A is the smallest (with respect to inclusion) subgroup of G containing A , or equivalently the set of elements of G that can be expressed as a finite product of elements in $\{a, a^{-1} : a \in A\}$ (i.e. take $\{a, a^{-1} : a \in A\}$ and shake vigorously).

To be precise,

$$\langle A \rangle = \bigcap_{\substack{H \supseteq A \\ H \leq G}} H.$$

This is not too practical for concrete descriptions of generated subgroups, but still good to keep in mind.

Example 2.8

Given $g \in G$, the generated subgroup $\langle g \rangle$ is $\{g^n : n \in \mathbb{Z}\}$. If this set is infinite, then it is isomorphic to $\mathbb{Z}$ via the isomorphism $n \mapsto g^n$. In this case, we say that g has infinite order. Otherwise, $g^m = 1$ for some m ; if m is taken to be positive and as small as possible, then the kernel of the $n \mapsto g^n$ isomorphism has kernel $m\mathbb{Z} = \{mn : n \in \mathbb{Z}\}$, hence $\langle g \rangle \cong \mathbb{Z}/m\mathbb{Z}$ (which is the group of integers modulo m under addition). We say here that g has order m . In general we denote the order of $g \in G$ with $|g|$.

Theorem 2.9 (Lagrange). If G is finite and H is a subgroup of G , then $|H|$ divides $|G|$.

Proof. For $a, b \in G$, say that $a \sim b$ if $a^{-1}b \in H$. It is easy to see that this an equivalence relation due to H being a subgroup of G . For a fixed a , then, the equivalence class of a

$$\{b \in G : b \sim a\} = \{ah : h \in H\} =: aH$$

(also called a **left coset** of H) has the same size as H , hence $|G| = |H| \cdot (\# \text{ of left cosets of } H)$ so $|G|/|H|$ is an integer. $\square$

The number of left cosets of H in G is called the **index** of H in G and is denoted $(G : H)$; the proof above shows that if G is finite then $(G : H) = |G| / |H|$.

Another important homomorphism is the **sign homomorphism** $\text{sgn} : S_n \rightarrow \{\pm 1\}$. Noting that S_n is generated by the set of 2-cycles (**transpositions**), we say that $\text{sgn}((a\ b)) = -1$ for all $a \neq b$ in $[n]$ and then extend multiplicatively (so for example 3-cycles, which are always a product of two 2-cycles, all return 1 when passed through the sgn homomorphism). The permutations σ with $\text{sgn}(\sigma) = -1$ are called odd permutations and those with $\text{sgn}(\sigma) = 1$ are even. The even permutations, being the kernel of sgn , form a normal subgroup of S_n called the **alternating group** A_n .

3 September 3

If $N \trianglelefteq G$ then

$$Nb = bN \implies aN \cdot bN = abN,$$

i.e. multiplication of cosets is well-defined here.

Definition 3.1 (Quotient group) — Given $N \trianglelefteq G$, the **quotient group** or **factor group** G/N is the group of left cosets of N with the multiplication $aN \cdot bN = abN$.

Note that Lagrange's theorem tells us $|G/N| = (G : N)$, the index of N in G . Additionally, there is a natural homomorphism $\pi : G \rightarrow G/N$ mapping

$$\pi(g) = gN,$$

whose kernel is $\{g \in G : gN = N\} = N$, so every normal subgroup of G is the kernel of some homomorphism out of G .

Now we turn to the four isomorphism theorems, which tell us about the structure of a group's subgroups in terms of homomorphisms going out of the group.

Theorem 3.2 (First isomorphism theorem). Let $\varphi : G \rightarrow H$ be a group homomorphism with kernel K . Then, $K \trianglelefteq G$ and $\varphi(G) = \overline{G}$ is a subgroup of H isomorphic to G/K with isomorphism $G/K \rightarrow \overline{G}$ given by $gK \mapsto \varphi(g)$.

This is a very simple and powerful theorem; proving that a group is isomorphic to a quotient often happens via the first isomorphism theorem.

Theorem 3.3 (Second isomorphism theorem). Let G be a group and $H \leq G$, $K \trianglelefteq G$.

- (1) $HK = \{h \cdot k : h \in H, k \in K\}$ is a subgroup of G .
- (2) $H \cap K \trianglelefteq H$.
- (3) $HK/K \cong H/(H \cap K)$.

Example 3.4

Consider the normal subgroup $K = \{(1), (1\ 2)(3\ 4), (1\ 3)(2\ 4), (1\ 4)(2\ 3)\}$ of S_4 and H the subgroup of permutations that fix 4, so $H \cong S_3$. Then, $H \cap K = \{1\}$, so by [Theorem 3.3](#)

$$HK/K \cong H/H \cap K = H \cong S_3.$$

This implies that $|HK| = |K| \cdot |S_3| = 24$, so in fact $HK = S_4$.

The third and fourth isomorphism theorems are very closely related:

Theorem 3.5 (Third isomorphism theorem). Let G be a group and $N \trianglelefteq G$. If $K \trianglelefteq G$ is such that $N \leq K \leq G$ then

- (1) $K/N \trianglelefteq G/N$
- (2) $(G/N)/(K/N) \cong G/K$.

This one says that normal subgroups K “pass through” quotients as long as K contains the group N that G is being quotiented by.

Theorem 3.6 (Fourth isomorphism theorem). Suppose G is a group and $N \trianglelefteq G$. There is a 1-1 correspondence between subgroups $H \leq G$ containing N and subgroups of G/N by the mapping $H \mapsto H/N$.

The fourth isomorphism theorem says that in fact the subgroups which pass through the quotient G/N (in the same spirit as in [Theorem 3.5](#)) are all the subgroups of G/N . If one were to look at the subgroups of G as a poset ordered by inclusion, the fourth isomorphism theorem says that the corresponding poset for G/N looks like zooming in on just the section of the poset bigger than N and smaller than G .

Thus, if $N \trianglelefteq G$, then we can try to understand the structure of G by examining

- (1) the smaller groups N and G/N . Repeating this analysis eventually reduces to the case where G has no nontrivial normal subgroups, where G is called **simple**.
- (2) how G is put together from N and G/N .

4 September 8

Two ways to study a group:

- (1) For a field F consider $GL_n(F)$. We can study G by homomorphisms $\rho: G \rightarrow GL_n(F)$; this is called **representation theory** (not discussed in this class)..
- (2) For a set X , we can consider the symmetric group $S_X = \{f: X \rightarrow X : f \text{ is bijective}\}$. Then, we can study G by homomorphisms $\rho: G \rightarrow S_X$. In this case we say that G **acts on** X or that G defines a **group action** on X .

We shall elaborate on the latter point.

Definition 4.1 (Group action) — For each $g \in G$, we associate a bijection $\rho(g): X \rightarrow X$ and write $g \cdot x := \rho(g)(x)$ such that the following hold:

- (1) $\rho(gh) = \rho(g) \circ \rho(h)$
- (2) $\rho(1) = \text{id}_X$, so $1 \cdot x = x$ for all x .

As mentioned above, we say that G **acts on** the set X . Since the group action is written to the left here, this is called a **left action** of G on X .

Example 4.2

Some examples of group actions:

- (1) D_8 acts on the corners of a square, as per its definition. If one were to label the vertices 1 through 4, we would get a homomorphism $D_8 \rightarrow S_4$ (see example [REFERENCE, 2.2]).
- (2) If G is a group and H a subgroup, then G acts on the set of all left cosets of H by $g \cdot aH = gaH$; it is an action because multiplication in G is associative. If $(G : H)$ is finite then this gives a homomorphism $G \rightarrow S_n$.
- (3) A group G can act on itself via conjugation, namely $g \cdot h = ghg^{-1}$.

Definition 4.3 (Orbits and stabilizers) — Suppose G acts on a set X . For a given $x \in X$, the **orbit** of x is the set $G \cdot x = \{g \cdot x : g \in G\}$. An element $x \in X$ is a **fixed point** if it is fixed by all $g \in G$, i.e. $g \cdot x = x$ for all g . G acts **transitively** on X if all of X is one orbit. Given $x \in X$, the **stabilizer** of x is the subgroup of G that fixes x , namely

$$G_x := \{g \in G : g \cdot x = x\}.$$

Exercise 4.4. Check that “being in the same orbit” is an equivalence relation on X , hence the orbits (equivalence classes) partition X .

Exercise 4.5. Show that for any $x \in X$, $G_x \leq G$.

Example 4.6 (A famous example)

Let $\mathbb{H} = \{z \in \mathbb{C} : \text{Im}(z) > 0\}$ be the upper half plane and $G = SL_2(\mathbb{R})$. Then, G acts on the Riemann sphere $\hat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$ by

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot z = \frac{az + b}{cz + d}.$$

Verifying that this is an action is a simple calculation.

The orbit of 0 is the entire real line $\mathbb{R} \cup \{\infty\}$.

If $y \in G \cdot x$, say $y = g \cdot x$, then

$$h \in G_y \implies h \cdot y = y \implies h \cdot g \cdot x = g \cdot x \implies (g^{-1}hg) \cdot x = x,$$

so if x is in the orbit of y then elements of the stabilizer of y are conjugate to elements of the stabilizer of x ; specifically, $G_y = g \cdot G_x \cdot g^{-1}$.

Consider $f: G \rightarrow G \cdot x$ by $g \mapsto g \cdot x$. Obviously f is surjective. Moreover, consider

$$f^{-1}(y) = \{g \in G : g \cdot x = y\}.$$

If $g \cdot x = h \cdot x$, then $h^{-1}g \cdot x = x$, meaning $h^{-1}g \in G_x$ or equivalently g is in the left coset $h \cdot G_x$. This means $f^{-1}(y)$ is a left coset of G_x , hence left cosets of G_x are in bijection with the orbit $G \cdot x$ via $gG_x \mapsto g \cdot x$. In the case where the orbit is finite, the size of the orbit is $|G \cdot x| = (G : G_x)$. When G is a finite group acting on a set X , we thus have

$$|G| = |G \cdot x| \cdot |G_x|.$$

When X is also finite, we have X as the disjoint union $\bigsqcup G \cdot x_i$ where we choose one representative x_i of each orbit, so

$$|X| = \sum_i |G \cdot x_i| = \sum_i \frac{|G|}{|G_{x_i}|}.$$

An important special case is the conjugation action described in [REFERENCE whichever example i wrote about conjugation in], namely $\alpha: G \times G \rightarrow G$ by $\alpha(g, h) = ghg^{-1}$. The orbit of an $h \in G$ is the **conjugacy class** $\{ghg^{-1} : g \in G\}$ of h . The stabilizer of h is

$$G_h = \{g \in G : ghg^{-1} = h \iff gh = hg\},$$

i.e. the elements in G that commute with h . This set is called the **centralizer** $C(h)$ of h . Of course, this means we can write the center of G as

$$Z(G) = \bigcap_{h \in G} C(h).$$

Exercise 4.7. When is $h \in G$ a fixed point under conjugation?

If G is finite and written as the union of conjugacy classes of h_i (so we pick one h_i from each conjugacy class)

$$|G| = \sum_i |\text{conjugacy class of } h_i| = \sum_i \frac{|G|}{|C(h_i)|} = |Z(G)| + \sum_{C(h_j) \neq G} \frac{|G|}{|C(h_j)|}$$

where the last expression groups together all the singleton orbits and so the summation runs over representatives h_j of orbits with size greater than 1 (i.e. $h_j \notin Z(G)$). This is called the **class equation** of G . The neat thing is that analyzing the addends in this equation can give constraints on the group because each addend must be a factor of $|G|$.